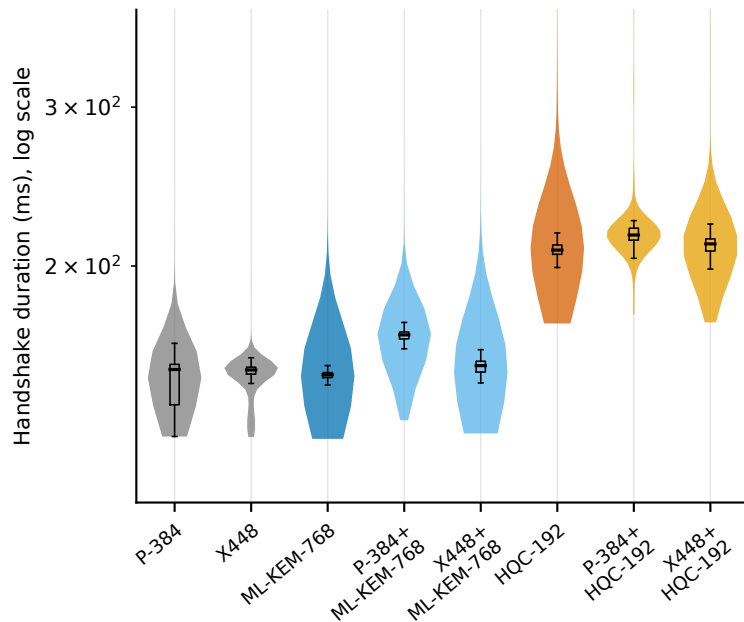
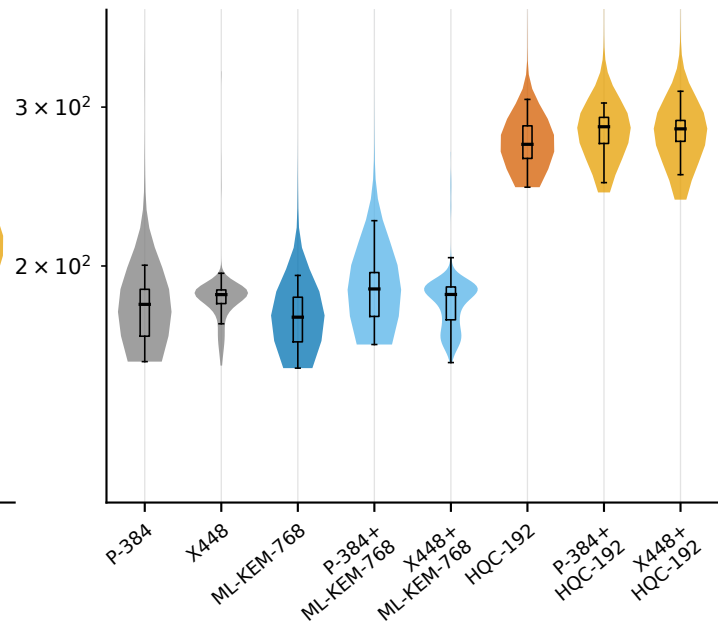


**(a)**

secp384r1 (classical)

**(b)**

mldsa65 (post-quantum)

**(c)**

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM              | secp384r1<br>Mean (ms) | secp384r1<br>CV (%) | secp384r1<br>Max (ms) | mldsa65<br>Mean (ms) | mldsa65<br>CV (%) | mldsa65<br>Max (ms) |
|------------------|------------------------|---------------------|-----------------------|----------------------|-------------------|---------------------|
| P-384            | 151.2                  | 30.6                | 1.16k                 | 183.6                | 36.1              | 1.21k               |
| X448             | 151.2                  | 9.0                 | 403.8                 | 184.5                | 7.2               | 328.7               |
| ML-KEM-768       | 156.7                  | 46.1                | 1.23k                 | 177.5                | 28.0              | 1.25k               |
| P-384+ML-KEM-768 | 170.6                  | 31.7                | 1.23k                 | 192.4                | 40.0              | 1.38k               |
| X448+ML-KEM-768  | 160.3                  | 50.4                | 1.22k                 | 183.0                | 7.1               | 267.6               |
| HQC-192          | 219.8                  | 46.6                | 1.39k                 | 276.8                | 20.4              | 1.40k               |
| P-384+HQC-192    | 219.6                  | 12.1                | 522.0                 | 287.3                | 22.9              | 1.31k               |
| X448+HQC-192     | 217.8                  | 32.2                | 1.23k                 | 287.9                | 25.7              | 1.40k               |